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UPI Transaction Analysis

Executive Report - Strategic Insights & Recommendations

Total Transactions	Transaction Amount	Avg. Amount	Failure Rate	Fraud Rate
100,000	₹4.24M	₹42.42	5.87%	2.00%

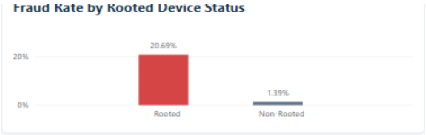
Executive Summary

Analysis of 100,000 UPI transactions identified an overall fraud rate of 2.00% and a transaction failure rate of 5.87%. The strongest validated risk signal was device rooting: rooted-device transactions recorded a 20.69% fraud rate versus 1.39% for non-rooted devices. Operational failures were spread across four major causes, while 87.60% of fraud alerts were resolved. Statistical testing did not support transaction channel as a significant fraud driver, so interventions should prioritize evidence-backed device risk and operational reliability.

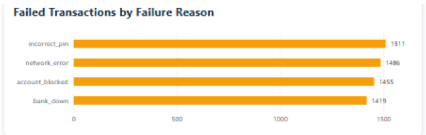
Top 5 Data-Driven Insights

Insight	Evidence	Business interpretation
Rooted devices are the clearest fraud-risk segment.	3,171 rooted-device transactions produced 656 fraud flags. Fraud rate: 20.69% vs 1.39% non-rooted.	Rooted devices are only 3.17% of volume but 32.8% of fraud flags; observed fraud rate is ~14.9x higher.
Transaction failures are distributed across multiple causes.	5,871 failures: incorrect PIN 1,511; network error 1,486; account blocked 1,455; bank down 1,419.	No single cause dominates; reliability improvement needs a multi-cause programme.
Fraud-alert resolution is strong, with a remaining backlog.	2,000 alerts: 1,752 resolved (87.60%) and 248 unresolved (12.40%).	Alert operations resolve most cases, but the unresolved queue remains material.
Channel fraud-rate differences are small and not statistically significant.	Observed rates: QR 2.10%, app 2.02%, intent 1.89%; Step-7 chi-square p = 0.143.	This analysis does not justify channel-specific restrictions based on fraud rate alone.
Entity fraud ratios require denominator context.	Some device ratios reach 100% on only 1 transaction; merchant rankings can also have small counts.	Use fraud ratio together with transaction count and fraud-flag count before escalation.

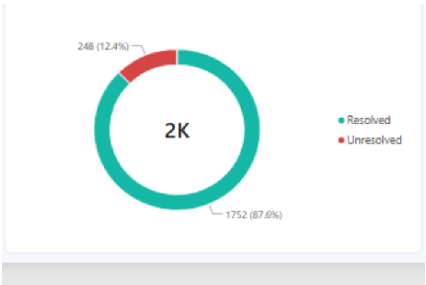
Supporting Dashboard Evidence



Rooted-device fraud comparison



Failure root-cause distribution



Fraud-alert resolution status

Strategic Recommendations & Expected Impact

Decision actions grounded in observed project metrics

Impact-estimate note: The percentages below are illustrative scenarios based on observed metrics. They are not forecasts or causal predictions.

Recommended Actions

#	Recommendation	Metric basis	Action	Illustrative impact
1	**Strengthen controls for rooted devices**	20.69% rooted fraud rate vs 1.39% non-rooted; 656 rooted-device fraud flags.	Use device-integrity checks, step-up authentication, additional verification and enhanced monitoring for rooted devices.	If rooted-device fraud flags fell by 25%, about 164 flags would be avoided; total fraud flags would fall from 2,000 to ~1,836 and overall observed fraud rate to ~1.84%.
2	**Run a multi-cause transaction-reliability programme**	5,871 failures; four causes each contribute roughly 24-26%.	Combine PIN guidance, network retry/resilience, blocked-account review and bank-availability monitoring.	A 10% reduction in current failures would remove ~587 failures, reducing failures to ~5,284 and the failure rate to ~5.28%.
3	**Reduce the unresolved fraud-alert backlog**	248 unresolved alerts = 12.40% of the 2,000-alert queue.	Introduce alert ageing, queue prioritization, ownership/SLA tracking and regular unresolved-alert review.	Resolving half the current backlog would reduce unresolved alerts to 124 and raise the observed resolution rate from 87.60% to ~93.80%.
4	**Avoid unsupported channel-specific restrictions**	Channel rates are close and Step-7 chi-square p = 0.143.	Maintain cross-channel monitoring rather than imposing additional controls solely on QR, app or intent transactions.	Helps avoid diverting security effort toward a channel for which this analysis does not demonstrate materially higher fraud risk.
5	**Add exposure context to device and merchant monitoring**	Very high fraud ratios can occur on tiny transaction counts.	Review Fraud Ratio + Transactions + Fraud-Flagged Transactions together before escalation; do not infer a fixed risk threshold.	Improves prioritization by reducing the chance that tiny-volume entities are escalated only because their percentage looks extreme.

Priority Framework

Immediate priority	Operational priority	Control priority	Monitoring principle
Rooted-device controls	**Failure reduction**	**Unresolved-alert management**	**Evidence-based entity/channel review**

Executive decision focus: prioritize the statistically supported device-risk signal, improve multi-cause transaction reliability, and strengthen alert-queue closure while preserving evidence-based monitoring.